

Molecular Mechanisms Driving Polysaccharide Enrichment in Sea Spray Aerosol Proxy Systems

Sea spray aerosol (SSA) is one of the most abundant sources of natural aerosol with significant but poorly understood impacts on Earth’s radiative energy budget.^{1,2} Organic compounds produced by marine biota partition to the air-sea interface, known as the sea surface microlayer (SSML), where the organic matter is transferred to SSA upon wave breaking.³ These organic molecules form a film around the aqueous salty core of the particle.⁴ Although sugars are generally regarded as water soluble and surface inactive, sugars have been measured with high enrichment factors in SSA as compared to seawater.^{5,6} To determine the molecular mechanisms behind sugar enrichment in SSA, a proxy SSML system was developed. SSA is coated in highly surface-active aliphatic compounds, with palmitic acid being the most abundant surfactant in SSA and the SSML.⁷ Na^+ , Ca^{2+} , and Cl^- comprise three of the most prevalent ions in seawater.^{8,9} Alginate serves as a marine-relevant model polysaccharide that readily forms gels in the presence of Ca^{2+} .^{10,11,12,13} The polysaccharide was enriched in laboratory-generated SSA, and its enrichment factor was higher in the presence of CaCl_2 .⁷ Consequently, the molecular organization of palmitic acid monolayers on the surface of NaCl aqueous solutions was monitored for changes upon introducing alginate and alginate- Ca^{2+} gels into the system. Changes in film molecular organization yield information regarding the potential mechanisms of polysaccharide enrichment in SSA.

Multiple surface-sensitive techniques were used to study changes in palmitic acid film organization upon the addition of alginate and CaCl_2 . Pendant drop surface tensiometry measured equilibrium surface tension of palmitic-acid-coated salty water droplets with alginate and alginate- Ca^{2+} gels within the droplet. Infrared reflection-absorption spectroscopy (IRRAS) and vibrational sum frequency generation spectroscopy (VSFG) probed interfacial molecular organization via functional group vibrational signatures. More specifically, IRRAS was used to determine palmitic acid lattice packing at the air-water interface as a function of Ca^{2+} and alginate presence.⁷ VSFG was used to determine palmitic acid alkyl chain orientation.⁷ The surface propensity of alginate was also confirmed by IRRAS and VSFG spectra. Lastly, Brewster angle microscopy (BAM) was used to visualize changes in film morphology after introducing alginate and Ca^{2+} to the system.⁷ Alginate was surface active, as demonstrated by the small surface tension depression of the fatty-acid-coated salty water droplet. Additionally, alginate disordered the film and increased the lipid tilt angle, indicating that the polysaccharide interacts with the palmitic acid monolayer. BAM images revealed that a small amount of alginate intercalated between surfactants in the monolayer; however, the majority of the alginate likely adsorbed to the palmitic acid head groups immersed in the aqueous phase of the system. Adding CaCl_2 produced greater surface tension depression and more gel particles visible at the surface. Larger palmitic acid tilt angles and significant monolayer disorder were also observed. Alginate- Ca^{2+} gels are likely highly surface active and intercalate into the monolayer due to their more compact structures. Alginate in the absence of calcium does not form such tight gels, so it is generally too large to intercalate between the monolayer lipids and simply adsorbs to the head groups of the film. Therefore, it is expected that intermolecular interactions between the polysaccharide gels and fatty acid films and gel intercalation into the SSML likely account for the high sugar enrichment in SSA.